

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

1	Integrals	2
1.1	Medium Integral	2
1.2	Hard Integral	3
2	Differentials	4
2.1	Hard Differential	4
3	Derivatives	5
3.1	Medium Derivative	5
3.2	Hard Derivative	5
4	Physics & Engineering	7
4.1	Astrophysics	7
4.2	Quantum Mechanics	7
4.3	Electromagnetism	8

Integrals

1.1 Medium Integral

Trigonometric Rational Integral

Evaluate the integral:

$$\int_{\pi/4}^{3\pi/4} \frac{1}{\sin^2(x) + \tan^2(x)} dx$$

Solution: First, rewrite the integrand in terms of sine and cosine:

$$\frac{1}{\sin^2(x) + \frac{\sin^2(x)}{\cos^2(x)}} = \frac{\cos^2(x)}{\sin^2(x)(\cos^2(x) + 1)}$$

To simplify the integration, we use the substitution $u = \cot(x)$. The differential is $du = -\csc^2(x) dx = -(1 + u^2) dx$, giving $dx = -\frac{du}{1+u^2}$. We also express the squared trigonometric functions in terms of u :

$$\sin^2(x) = \frac{1}{1+u^2}, \quad \cos^2(x) = \frac{u^2}{1+u^2}$$

The integration limits change from $x = \pi/4 \implies u = 1$ to $x = 3\pi/4 \implies u = -1$. Substituting these:

$$\int_1^{-1} \frac{\frac{u^2}{1+u^2}}{\frac{1}{1+u^2} \left(\frac{u^2}{1+u^2} + 1 \right)} \left(\frac{-1}{1+u^2} \right) du$$

Simplify the denominator of the main fraction:

$$\frac{u^2}{1+u^2} + 1 = \frac{2u^2 + 1}{1+u^2}$$

The integral condenses to:

$$\int_1^{-1} \frac{u^2}{\frac{2u^2+1}{1+u^2}} \left(\frac{-1}{1+u^2} \right) du = \int_{-1}^1 \frac{u^2}{2u^2+1} du$$

Using algebraic manipulation on the numerator:

$$\frac{1}{2} \int_{-1}^1 \frac{2u^2 + 1 - 1}{2u^2 + 1} du = \frac{1}{2} \int_{-1}^1 \left(1 - \frac{1}{2u^2 + 1} \right) du$$

Integrate term by term:

$$\frac{1}{2} \left[u - \frac{1}{\sqrt{2}} \arctan(u\sqrt{2}) \right]_{-1}^1$$

Plug in the upper and lower limits:

$$\frac{1}{2} \left(\left(1 - \frac{\arctan(\sqrt{2})}{\sqrt{2}} \right) - \left(-1 - \frac{\arctan(-\sqrt{2})}{\sqrt{2}} \right) \right) = 1 - \frac{\arctan(\sqrt{2})}{\sqrt{2}}$$

1.2 Hard Integral

Logarithmic and Inverse Tangent Integral

Evaluate the integral:

$$\int_0^1 \frac{\tan^{-1}(x)}{x} \ln \left(\frac{1+x^2}{(1-x)^2} \right) dx$$

Solution: Let us apply the trigonometric substitution $x = \tan(\theta)$. The integral transforms into:

$$\int_0^{\pi/4} \frac{\theta}{\tan(\theta)} \ln \left(\frac{\sec^2(\theta)}{(1-\tan(\theta))^2} \right) \sec^2(\theta) d\theta$$

Simplify the argument inside the natural logarithm:

$$\frac{\sec^2(\theta)}{(1-\tan(\theta))^2} = \frac{1}{\cos^2(\theta) \left(1 - \frac{\sin(\theta)}{\cos(\theta)}\right)^2} = \frac{1}{(\cos(\theta) - \sin(\theta))^2}$$

Thus, the integral is rewritten as:

$$\int_0^{\pi/4} \frac{\theta}{\sin(\theta) \cos(\theta)} \ln \left((\cos(\theta) - \sin(\theta))^{-2} \right) d\theta$$

Applying the double angle identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$ and bringing the -2 exponent to the front:

$$-4 \int_0^{\pi/4} \frac{\theta}{\sin(2\theta)} \ln(\cos(\theta) - \sin(\theta)) d\theta$$

Let $\phi = 2\theta$, yielding $d\theta = \frac{1}{2}d\phi$. The bounds shift to $[0, \pi/2]$:

$$-4 \int_0^{\pi/2} \frac{\phi/2}{\sin(\phi)} \ln \left(\cos \left(\frac{\phi}{2} \right) - \sin \left(\frac{\phi}{2} \right) \right) \frac{1}{2} d\phi = - \int_0^{\pi/2} \frac{\phi}{\sin(\phi)} \ln \left(\cos \left(\frac{\phi}{2} \right) - \sin \left(\frac{\phi}{2} \right) \right) d\phi$$

Examine the square of the logarithmic argument:

$$\left(\cos \left(\frac{\phi}{2} \right) - \sin \left(\frac{\phi}{2} \right) \right)^2 = \cos^2 \left(\frac{\phi}{2} \right) + \sin^2 \left(\frac{\phi}{2} \right) - 2 \sin \left(\frac{\phi}{2} \right) \cos \left(\frac{\phi}{2} \right) = 1 - \sin(\phi)$$

This transforms the logarithm into $\frac{1}{2} \ln(1 - \sin(\phi))$. Substituting this yields:

$$-\frac{1}{2} \int_0^{\pi/2} \frac{\phi \ln(1 - \sin(\phi))}{\sin(\phi)} d\phi$$

The remaining definite integral is a known result which evaluates precisely to $-\frac{\pi^3}{8}$. Multiplying by the constant:

$$-\frac{1}{2} \left(-\frac{\pi^3}{8} \right) = \frac{\pi^3}{16}$$

Differentials

2.1 Hard Differential

Exact Equation and Integrating Factor

Solve the differential equation:

$$x dy - y dx = (x^2 + y^2) dx$$

Given the initial condition $y(1) = 0$, find the exact value of $y\left(1 + \frac{\pi}{4}\right)$.

Solution: Divide both sides by $(x^2 + y^2)$:

$$\frac{x dy - y dx}{x^2 + y^2} = dx$$

Recognize the left side as the exact differential of the arctangent function. Using the quotient rule on $\arctan(y/x)$ gives:

$$d\left(\arctan\left(\frac{y}{x}\right)\right) = \frac{1}{1 + (y/x)^2} \left(\frac{x dy - y dx}{x^2}\right) = \frac{x dy - y dx}{x^2 + y^2}$$

Substituting this back into the ODE:

$$d\left(\arctan\left(\frac{y}{x}\right)\right) = dx$$

Integrating both sides yields:

$$\arctan\left(\frac{y}{x}\right) = x + C$$

Applying the initial condition $y(1) = 0$:

$$\arctan\left(\frac{0}{1}\right) = 1 + C \implies C = -1$$

The particular solution is:

$$\arctan\left(\frac{y}{x}\right) = x - 1 \implies \frac{y}{x} = \tan(x - 1) \implies y = x \tan(x - 1)$$

To find y evaluated at $x = 1 + \frac{\pi}{4}$:

$$y\left(1 + \frac{\pi}{4}\right) = \left(1 + \frac{\pi}{4}\right) \tan\left(1 + \frac{\pi}{4} - 1\right) = \left(1 + \frac{\pi}{4}\right) \tan\left(\frac{\pi}{4}\right)$$

Because $\tan(\pi/4) = 1$, the exact value is:

$$1 + \frac{\pi}{4}$$

Derivatives

3.1 Medium Derivative

Tangent Line and Triangle Area

Let $f(x) = x^2 + 4x + 5$ and let $P = (a, f(a))$ with $a > 0$. The tangent line to the curve at P intercepts the x -axis at A and the y -axis at B . If the area of triangle AOB is 4, find a . Note: $O = (0, 0)$.

Solution: First, take the derivative of $f(x)$ to establish the slope of the tangent line:

$$f'(x) = 2x + 4$$

At point $P(a, a^2 + 4a + 5)$, the slope is $m = 2a + 4$. The equation of the tangent line becomes:

$$y - (a^2 + 4a + 5) = (2a + 4)(x - a)$$

Set $x = 0$ to find the y -intercept B :

$$y = (2a + 4)(-a) + a^2 + 4a + 5 = -2a^2 - 4a + a^2 + 4a + 5 = 5 - a^2$$

So $B = (0, 5 - a^2)$. Next, set $y = 0$ to find the x -intercept A :

$$\begin{aligned} -(a^2 + 4a + 5) &= (2a + 4)(x - a) \implies x - a = -\frac{a^2 + 4a + 5}{2a + 4} \\ x &= a - \frac{a^2 + 4a + 5}{2a + 4} = \frac{2a^2 + 4a - a^2 - 4a - 5}{2a + 4} = \frac{a^2 - 5}{2a + 4} \end{aligned}$$

So $A = \left(\frac{a^2 - 5}{2a + 4}, 0\right)$. The area of the right triangle AOB is $Area = \frac{1}{2}|x_A||y_B|$:

$$\frac{1}{2} \left| \frac{a^2 - 5}{2a + 4} \right| |5 - a^2| = \frac{(a^2 - 5)^2}{4(a + 2)}$$

Given $a > 0$, $(a + 2)$ is strictly positive. Equating the area expression to 4:

$$\frac{(a^2 - 5)^2}{4a + 8} = 4 \implies (a^2 - 5)^2 = 16a + 32 \implies a^4 - 10a^2 - 16a - 7 = 0$$

Testing integer roots shows $a = -1$ is a solution. Factoring out $(a + 1)$ reveals $a^3 - a^2 - 9a - 7 = 0$, where $a = -1$ is a root again. Fully factored:

$$(a + 1)^2(a^2 - 2a - 7) = 0$$

Since $a > 0$, we disregard $a = -1$ and apply the quadratic formula to the remaining factor:

$$a = \frac{2 \pm \sqrt{4 - 4(1)(-7)}}{2} = 1 \pm \sqrt{8} = 1 \pm 2\sqrt{2}$$

Taking the positive root, we conclude:

$$a = 1 + 2\sqrt{2}$$

3.2 Hard Derivative

Schwarzian Derivative Series

We define the Schwarzian Derivative:

$$D_S[f(x)] = \frac{f'''(x)}{f'(x)} - \frac{3}{2} \left[\frac{f''(x)}{f'(x)} \right]^2$$

Compute:

$$\sum_{n=2}^{\infty} \frac{1}{2^n} \cdot D_S \left[\frac{ne^x + n + 1}{n^2e^x + n^2 + 1} \right]$$

Solution: A core property of the Schwarzian derivative is its invariance under non-degenerate fractional linear (Möbius) transformations. For constants A, B, C, D where $AD - BC \neq 0$:

$$D_S \left[\frac{A \cdot g(x) + B}{C \cdot g(x) + D} \right] = D_S[g(x)]$$

Here, we set $g(x) = e^x$. The given rational function takes the form $\frac{Ae^x+B}{Ce^x+D}$ using $A = n$, $B = n + 1$, $C = n^2$, and $D = n^2 + 1$. Checking the determinant:

$$AD - BC = n(n^2 + 1) - n^2(n + 1) = n^3 + n - n^3 - n^2 = n - n^2$$

Because $n \geq 2$, the determinant $n - n^2 \neq 0$, meaning the transformation is indeed non-degenerate. The Schwarzian derivative reduces purely to $D_S[e^x]$. Because $f(x) = f'(x) = f''(x) = f'''(x) = e^x$, we get:

$$D_S[e^x] = \frac{e^x}{e^x} - \frac{3}{2} \left(\frac{e^x}{e^x} \right)^2 = 1 - \frac{3}{2} = -\frac{1}{2}$$

The infinite sum now translates to a simple geometric series:

$$\sum_{n=2}^{\infty} \frac{1}{2^n} \left(-\frac{1}{2} \right) = -\frac{1}{2} \sum_{n=2}^{\infty} \left(\frac{1}{2} \right)^n$$

Evaluating the geometric series:

$$\sum_{n=2}^{\infty} \left(\frac{1}{2} \right)^n = \frac{(1/2)^2}{1 - 1/2} = \frac{1/4}{1/2} = \frac{1}{2}$$

Multiplying by our extracted constant:

$$-\frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{4}$$

Physics & Engineering

4.1 Astrophysics

Total Space Velocity of Hamal

The star Hamal (α Arietis) is located $d = 20.2$ pc from Earth. It has a proper motion of $\mu = 190$ mas yr^{-1} and a radial velocity of $v_r = -14.9$ km s^{-1} . Calculate the magnitude of Hamal's total space velocity v (in km s^{-1}) relative to the Sun.

Solution: To find the total space velocity, first compute the transverse velocity v_t . Using the standard relation $v_t \approx 4.74 \cdot \mu \cdot d$ (where μ is proper motion in arcseconds per year and d is distance in parsecs):

$$\mu = 190 \text{ mas yr}^{-1} = 0.190 \text{ arcsec yr}^{-1}$$

$$v_t = 4.74 \times 0.190 \times 20.2 \approx 18.192 \text{ km s}^{-1}$$

The total space velocity v represents the vector sum of both transverse and radial velocities:

$$v = \sqrt{v_r^2 + v_t^2}$$

Substituting in the values:

$$v = \sqrt{(-14.9)^2 + (18.192)^2} = \sqrt{222.01 + 330.95} = \sqrt{552.96} \approx 23.5 \text{ km s}^{-1}$$

4.2 Quantum Mechanics

Stern-Gerlach Spin Probabilities

A beam of silver atoms is prepared in the spin state:

$$|\psi\rangle = \frac{4}{5}|\uparrow_z\rangle + \frac{3}{5}|\downarrow_z\rangle$$

The beam passes through a Stern-Gerlach apparatus measuring spin along the x -axis. Only atoms with eigenvalue $+\hbar/2$ are collected. This filtered beam then enters a second apparatus measuring spin along the y -axis. Calculate the probability that an atom is measured to have a spin eigenvalue of $-\hbar/2$.

Solution: When the initial beam passes through the first apparatus (S_x) and is filtered for atoms with the $+\hbar/2$ eigenvalue, the resulting beam collapses into the $+x$ eigenstate, erasing the original preparation state $|\psi\rangle$. The state of the newly filtered beam is:

$$|\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$$

This beam subsequently enters the second apparatus (S_y), where we check the probability of detecting the $-\hbar/2$ eigenvalue, which corresponds to the state $|\downarrow_y\rangle$. The $-y$ eigenstate is:

$$|\downarrow_y\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle - i|\downarrow_z\rangle)$$

The probability of this outcome is the squared magnitude of the inner product $|\langle\downarrow_y|\uparrow_x\rangle|^2$. Calculating the inner product (and remembering that taking the bra $\langle\downarrow_y|$ conjugates $-i$ to $+i$):

$$\langle\downarrow_y|\uparrow_x\rangle = \frac{1}{\sqrt{2}}(\langle\uparrow_z| + i\langle\downarrow_z|) \cdot \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$$

$$\langle\downarrow_y|\uparrow_x\rangle = \frac{1}{2}(1 \cdot 1 + i \cdot 1) = \frac{1+i}{2}$$

Squaring the absolute magnitude of the amplitude gives the probability:

$$P = \left| \frac{1+i}{2} \right|^2 = \frac{1^2 + 1^2}{2^2} = \frac{2}{4} = \frac{1}{2}$$

4.3 Electromagnetism

Mass of a Helium Ion

A beam of singly ionised helium ions (He^+) enters a region where a uniform electric field and magnetic field are perpendicular to one another. The ions travel undeflected when the electric field strength is $E = 3.60 \times 10^4 \text{ V m}^{-1}$ and the magnetic flux density is $B = 0.240 \text{ T}$. The ions then enter a second region containing only the magnetic field, where they follow a circular path of radius $r = 0.518 \text{ m}$. Calculate the mass of a helium ion. Give the answer as a power of 10^{-25} to 2 decimal places. Use $e = 1.602 \times 10^{-19} \text{ C}$.

Solution: In the first region, which acts as a velocity selector, the electric and magnetic forces must perfectly balance for undeflected travel:

$$qE = qvB \implies v = \frac{E}{B}$$

Find the velocity by substituting the respective field strengths:

$$v = \frac{3.60 \times 10^4}{0.240} = 1.50 \times 10^5 \text{ m s}^{-1}$$

Upon entering the second region, the uniform magnetic field produces the centripetal force driving the circular motion:

$$qvB = \frac{mv^2}{r} \implies m = \frac{qBr}{v}$$

For a singly ionised atom, the charge $q = e = 1.602 \times 10^{-19}$ C. Substituting our parameters:

$$m = \frac{(1.602 \times 10^{-19} \text{ C}) \times (0.240 \text{ T}) \times (0.518 \text{ m})}{1.50 \times 10^5 \text{ m/s}}$$
$$m = \frac{0.19915264 \times 10^{-19}}{1.50 \times 10^5} = 0.1327684 \times 10^{-24} \text{ kg}$$

Converting to the requested scale:

$$m \approx 1.32768 \times 10^{-25} \text{ kg}$$

Rounding to 2 decimal places:

$$1.33 \times 10^{-25} \text{ kg}$$